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Primary Knee

Patient-Reported Anxiety or Depression Increased the Risk of Dissatisfaction Despite Improvement in Pain or Function Following Total Knee Arthroplasty: A Swedish Register-Based Observational Study of 8,745 Patients

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ABSTRACT

Background: Remaining pain and functional limitations may cause dissatisfaction in patients after total knee arthroplasty (TKA). Furthermore, anxiety and depression are associated with remaining postoperative symptoms and dissatisfaction. We investigated if patient-reported anxiety or depression increased the risk of dissatisfaction 1 year after TKA in patients who improved in pain or function.
Methods: Data on primary TKAs due to osteoarthritis between 2017 and 2019 were obtained from the Swedish Knee Arthroplasty Register. Out of 14,120 patients, 9,911 completed the Knee injury and Osteoarthritis Outcome Score and satisfaction rate with the result of the surgery. According to the Outcome Measures in Rheumatology—Osteoarthritis Research Society International criteria patients were classified as responders (improved in pain or function) or nonresponders. Anxiety and depression were assessed through EuroQol-5 Dimension 3 levels. Log-linear regression models estimated the risk ratios (RRs) for dissatisfaction in all patients and stratified by age groups (< 65, 65 to 74, and > 74 years). There were 8,745 patients who were classified as responders whereas 11% were defined as dissatisfied. The proportion of patients who reported anxiety or depression was 35% preoperatively and 17% postoperatively.
Results: Anxiety or depression increased the risk of dissatisfaction preoperatively (RR 1.23, 95% confidence interval = 1.09 to 1.40) and postoperatively (RR 2.65, confidence interval 2.33 to 3.00). Patients younger than 65 years reported preoperative anxiety or depression to a greater extent but did not have an increased risk of dissatisfaction.
Conclusions: Patient-reported anxiety or depression preoperatively and postoperatively are important and potentially treatable factors to consider, as they were found to increase the risk of dissatisfaction after TKA despite improvements in pain or function.
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Total knee arthroplasty (TKA) aims to relieve pain, improve function, and increase the health-related quality of life of patients who have advanced osteoarthritis (OA). In the last few decades, patient satisfaction has been given substantial attention as an indicator of treatment success. Most patients are satisfied with the result after TKA; however, there is a remaining number of patients who are not. According to a recent review, the dissatisfaction rate after TKA is approximately 10% [1], which is similar to the latest

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annual report from the Swedish Arthroplasty Register [2]. Given the aims of TKA, it is not surprising that anxiety or depression, persistent postoperative pain, and poor knee function are associated with dissatisfaction alongside other factors such as higher comorbidity, age, and unmet expectations [1,3–13].

A register-based study from a region in southern Sweden showed that individuals who were diagnosed with OA were more likely to be diagnosed with depression during the following years compared to individuals who did not have an OA diagnosis [14]. Although the prevalence of anxiety or depression decreases after a TKA [15–19], patients who have preoperative anxiety or depression have previously shown worse postoperative pain scores in patient-reported outcome measures (PROMs) and less improvement in their knee function [16,17,20]. Patients who are diagnosed with or have symptoms of anxiety or depression may be less able to engage in rehabilitation and have worse outcomes in terms of pain and function. Therefore, it is not clear if the risk of dissatisfaction in patients who have anxiety or depression is explained by potentially worse outcomes in pain and function or if the presence of anxiety or depression itself has relevance. This study was performed to reduce the confounding of no or less improvement in pain or function and other causes of dissatisfaction related to that by exclusively including TKA patients who were considered responders according to the Outcome Measures in Rheumatology—Osteoarthritis Research Society International (OMERACT-OARSI) responder criteria [21].

The primary aim was to investigate the risk of dissatisfaction 1 year after TKA in responders who had preoperative or postoperative patient-reported anxiety or depression. The secondary aim was to explore the aforementioned in different age groups.

Materials and Methods

Study Design, Data Collection, and Patient Selection

This was a register-based observational study with prospectively collected data assessed from the Swedish Knee Arthroplasty Register (SKAR) (the Swedish Arthroplasty Register since September 2021, http://sar.registercentrum.se). Ethical approval was obtained from the Faculty of Medicine ethics committee (Dnr. LU 20-02 and 331/2005). Demographical data and PROM were obtained from 17,219 TKAs, with OA as the primary diagnosis, performed between 2017 and 2019 at clinics that voluntarily participated in the SKAR PROM project [22]. Participating clinics in the SKAR PROM project were university clinics as well as public and private clinics geographically spread across Sweden. The PROMs were collected routinely preoperatively and 1 year postoperatively [22]. The PROMs assessed were the Knee injury and Osteoarthritis Outcome Score (https://eprovide.mapi-trust.org/), the EuroQol-5 Dimension 3 levels (EQ-5D 3L) (https://euroqol.org/), and satisfaction with the result of the surgery 1 year postoperatively. Demographical data included age, sex, body mass index (BMI), and the American Society of Anesthesiologists (ASA) classification. Exclusion criteria were contralateral TKA during follow-up (n = 1,870), bilateral TKA in 1 session (n = 1,170), death before follow-up (n = 59), and missing PROM data (n = 4,209) (Figure 1).

Responder Criteria

The OMERACT-OARSI responder criteria were used in the SKAR to determine if patients improved in pain or function 1 year

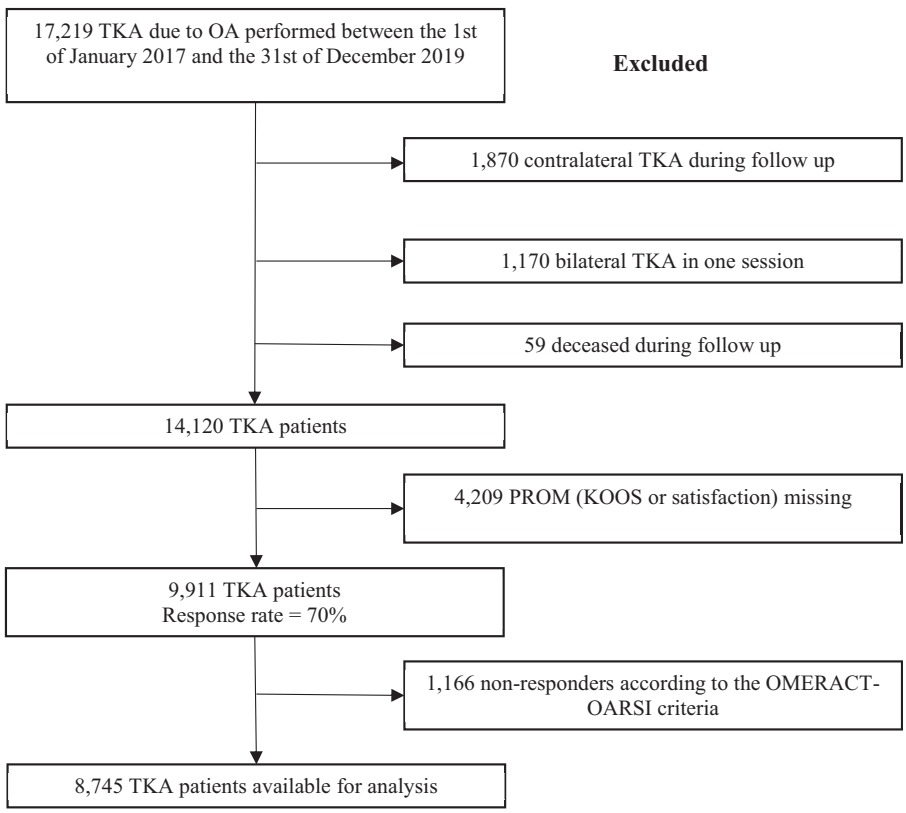

Fig. 1. Flow diagram of the study. TKA, total knee arthroplasty; OA, osteoarthritis; PROM, patient-reported outcome measures; KOOS, Knee injury and Osteoarthritis Outcome Score; OMERACT-OARSI, Outcome Measures in Rheumatology—Osteoarthritis Research Society International.

postoperatively [21]. The criteria assign patients to one of the following categories: high responders, low responders, or nonresponders. They are based on the absolute and relative changes between preoperative and 1-year postoperative knee-related function, pain, and stiffness assessed by the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) [23]. The Knee injury and Osteoarthritis Outcome Score was converted to WOMAC to determine the responder classification [22]. High responders were patients who had a relative improvement of at least 50% and an absolute improvement of at least 20 points in WOMAC pain or function. Low responders were patients who had a relative improvement of at least 20% and an absolute improvement of at least 10 points in 2 of either pain, function, or total score. The remaining patients were classified as nonresponders and were not included in the analyses (Figure 1).

Definition of Outcomes

Dissatisfaction was assessed with a single-item question asking patients about their satisfaction with the result of the surgery 1 year after TKA using a visual analog scale (VAS) ranging from highest imaginable satisfaction to highest imaginable dissatisfaction. We defined dissatisfaction as "not satisfied" and dichotomized the scores and labeled them satisfied patients (the best 40% of the VAS) and dissatisfied patients (the worst 60% of the VAS). Anxiety or depression was assessed from EQ-5D 3L preoperatively and 1 year postoperatively. Patients were asked to define their health state of anxiety or depression as none, moderate or extreme. We dichotomized the assessments to the presence (moderate or extreme) or absence of anxiety or depression from the EQ-5D 3L. Expectations of the surgery were assessed preoperatively with the question "Mark the place that you believe represents how well your knee will be after the surgery (compared to a healthy knee in your age)" using a VAS reaching from 0 (completely healthy) to 100 (worst imaginable).

Data Analyses

Demographical data are presented as mean values (standard deviations or ranges) and numbers (percentages). A multivariate log-linear regression analysis was performed to estimate the risk ratio (RRs) for dissatisfaction. The benefit of this approach is that the RR is easier to interpret than the odds ratio. Multiple analyses were carried out and referred to as models 1 and 2. The outcome variable was dissatisfaction 1 year after TKA. The exposure of interest in model 1 was preoperative patient-reported anxiety or depression and the exposure of interest in model 2 was patient-reported anxiety or depression 1 year after TKA. Stratified analyses were performed based on age: all patients, patients aged < 65 years, patients aged 65 to 74 years, and patients aged > 74 years. The analyses were adjusted for age groups, sex, expectations of surgery, and comorbidity (ASA classification) since they may impact the risk of dissatisfaction or anxiety/depression [1,24]. Estimates from the regression analyses are presented as RR (95% confidence intervals). The analyses were performed with SPSS Statistics 28.0.0.1 (IBM Corp. Armonk, NY) and R version 4.2.1 (R Foundation for Statistical Computing, Vienna, Austria; https://www.R-project.org/).

Patient Characteristics

Out of 9,911 patients, there were 8,745 who were classified as responders and included in the analyses (Figure 1). The mean age of the studied cohort was 70 years (range, 33 to 97), 58% were women, and the mean BMI was 29 (standard deviation = 4) (Table 1). There were 11% of the patients who were classified as dissatisfied.

Results

Preoperative and postoperative patient-reported anxiety or depression were associated with a higher risk of being dissatisfied 1 year after TKA (Table 2). This was observed across the models and age groups except for patients < 65 years who had preoperative anxiety or depression (Table 3). The proportion of patients who reported anxiety or depression was almost halved, from 35% at the preoperative stage to 17% 1 year after TKA (Table 1). Preoperative patient-reported anxiety or depression was more common in the youngest age group (41%), compared to the older age groups (32 and 35%); however, the postoperative proportions were similar in all groups (Table 1). The risk of dissatisfaction was higher in patients who reported anxiety or depression postoperatively compared to preoperatively (Tables 2 and 3). Lower expectations were associated with dissatisfaction in all analyses. Higher comorbidity (ASA class ≥ 3) was associated with dissatisfaction in the model including all age groups and the model including the subgroup of patients aged ≥ 74 years. Sex and age were not found to increase the risk of dissatisfaction 1 year after TKA. Missing values of patient-reported anxiety or depression were less than 1% (Table 1).

Characteristics of Patients who Had Missing PROMs

Patients who were excluded due to missing PROM data at follow-up (n = 4,209) (Figure 1) were similar in age, BMI, and sex; however, compared to the studied cohort, a higher proportion were classified as having ASA ≥ 3 (24 compared to 16%).

Table 1
Patient Characteristics, Dissatisfaction, and Presence of Anxiety or Depression in the Studied Cohort and Age Groups.

	All Patients n = 8,745	<65 Y n = 2,410 (28%)	65 to 74 Y n = 3,777 (43%)	>74 Y n = 2,558 (29%)
Mean age (range)	70 (33 to 97)			
Women, n (%)	5,079 (58)	1,450 (60)	2,164 (57)	1,465 (57)
Mean BMI (SD)	29 (4)	30 (4)	29 (4)	28 (4)
ASA classification ≥3, n (%)	1,369 (16)	225 (9)	561 (15)	583 (23)
Dissatisfaction, n (%)	926 (11)	267 (11)	394 (10)	265 (10)
Anxiety or depression, n (%)				
Present preoperatively	3,083 (35)	976 (41)	1,225 (32)	882 (35)
N/a preoperatively	46 (<1)	13 (<1)	16 (<1)	17 (<1)
Present postoperatively	1,442 (17)	444 (18)	547 (15)	451 (18)
N/a postoperatively	48 (<1)	10 (<1)	21 (<1)	17 (<1)

N, number; BMI, body mass index; SD, standard deviation; ASA, American Society of Anesthesiologists.

Table 2
Results From the Multivariate Log-Linear Regression Analyses at Being Dissatisfied 1 Y After Total Knee Arthroplasty.

All Age Groups (n = 8,745)	Model 1 Dissatisfaction		Model 2 Dissatisfaction	
	RR[a]	95% CI	RR	95% CI
Preoperative anxiety or depression				
No	ref	ref		
Yes	1.23	1.09 to 1.40		
Postoperative anxiety or depression				
No			ref	ref
Yes			2.65	2.33 to 3.00
ASA classification				
≤2	ref	ref	ref	ref
≥3	1.30	1.12 to 1.51	1.21	1.04 to 1.40
Women	ref	ref	ref	ref
Men	0.92	0.81 to 1.04	0.97	0.86 to 1.10
Expectations[b]	1.22	1.19 to 1.25	1.21	1.18 to 1.24
Age group				
<65 y	ref	ref	ref	ref
65 to 74 y	0.96	0.83 to 1.11	1.01	0.87 to 1.17
>74 y	0.88	0.75 to 1.03	0.88	0.75 to 1.03

The exposure of interest was preoperative anxiety or depression in Model 1 and postoperative anxiety or depression in Model 2.
ref, reference value; ASA, American Society of Anesthesiologists; RR, risk ratio; CI, confidence interval.
[a] Presented as risk ratio (RR) with 95% confidence interval (CI).
[b] For every 10 units increase in the visual analog scale ranging from 0 to 100 (high to low expectations).

Discussion

In this register-based observational study of OA patients who had TKA and were 1 year postoperatively classified as responders, we found that the risk of dissatisfaction was higher in patients who reported anxiety or depression. This was found when anxiety or depression was reported preoperatively and 1 year after TKA. Patients younger than 65 years reported preoperative anxiety or depression to a greater extent but did not have an increased risk of dissatisfaction at follow-up.

The results of this study provide further support that anxiety or depression are important, potentially treatable factors to consider in reducing the number of dissatisfied patients after TKA. Previous studies have shown that preoperative symptoms of anxiety or depression not only increase the risk of being dissatisfied [7,8] but also the risk of readmission [25], infection, reoperation, and revision surgery [26]. Our study was performed entirely on patients who were considered to have improved in pain or function, which are 2 important aims of TKA. The results indicate that anxiety or depression does not only act as a mediator of other worse outcomes related to pain or function that increases the risk of dissatisfaction. Several aspects related to anxiety or depression such as loss of energy, insomnia, or a depressed mood, could interfere with patients' ability to adapt to life with a knee prosthesis and be satisfied with the result after surgery. The first-line treatment offered for adult patients who have depression and anxiety in Sweden is cognitive behavior therapy, interpersonal psychotherapy, and serotonin reuptake inhibitors [24]. Patients who have symptoms of anxiety or depression but do not meet the criteria for a clinical diagnosis may be helped with other interventions preoperatively and postoperatively, such as patient empowerment or psychosocial support [27].

Interestingly, a study performed by Seagrave et al. in patients who underwent TKA and total hip arthroplasty showed that patients who had medical treatment for anxiety or depression did not have an increased risk of dissatisfaction [28]. Patients who are treated for anxiety or depression may be better equipped to undergo surgery. In the same way that patients who have, for example, diabetes mellitus are metabolically optimized before surgery to decrease the risk of complications. The proportion of patients in our study who reported anxiety or depression was reduced by 47% after the TKA. These findings follow those from previous studies [15–18]. There may be different natures of anxiety or depression in patients who have knee OA. Some of these may be triggered by symptomatology such as chronic knee pain or reduced quality of life and can therefore be relieved to some extent by TKA [15,18,19]. However, the risk of dissatisfaction was higher in patients who had anxiety or depression 1 year after TKA compared to preoperatively, indicating that patient-reported anxiety or depression that is not relieved by surgery may be an important risk factor for dissatisfaction. Since anxiety or depression have been shown to decrease after surgery, such symptoms are not necessarily a reason to postpone the TKA. However, it may be beneficial for patients if physicians assess symptoms of anxiety and depression preoperatively and postoperatively to identify patients who need psychological or pharmacological treatment and, as such, possibly reduce the number of dissatisfied patients and decrease the risk of complications.

Patients who were younger than 65 years reported anxiety or depression to a greater extent than elderly patients. There may be a larger stigma associated with OA among younger adults, more stress about the loss of income due to sick leave, family situations, and higher functional demands at work or in leisure and recreational activities. The postoperative proportions were similar in all groups, and younger patients reported, to a greater extent, no anxiety or depression after TKA. This was supported by the fact that patients younger than 65 years who reported preoperative anxiety/depression did not have an increased risk for dissatisfaction. In previous studies, the association between age and dissatisfaction after TKA was conflicting [1]. Younger patients' expectations of the surgical outcome may differ from those of elderly patients, who more frequently suffer from higher comorbidity, factors that have previously been shown to increase the risk of dissatisfaction [1].

In this study, 35% of the respondents reported preoperative anxiety or depression using the EQ-5D 3L. It must be emphasized that the meaning of the words used in the question that assesses anxiety or depression in the Swedish version of EQ-5D 3L is "oro" (anxiety), which if translated into English, refers to states such as worry or uneasiness. This word (eg, the word "oro" used in EQ-5D 3L) is different from the Swedish word for anxiety, such as used in generalized anxiety disorder or social anxiety disorder. Patients who express that they have "oro" preoperatively may, for example,

Table 3
Results From the Multivariate Log-Linear Regression Analyses at Being Dissatisfied 1 Y After Total Knee Arthroplasty Stratified by Age Groups.

< 65 y (n = 2,410)	Model 1 Dissatisfaction		Model 2 Dissatisfaction	
	RR[a]	95% CI	RR[a]	95% CI
Preoperative anxiety or depression				
No	ref			
Yes	1.15	0.91 to 1.44		
Postoperative anxiety or depression				
No			ref	
Yes			2.73	2.17 to 3.44
ASA class				
≤2	ref		ref	
≥3	0.89	0.60 to 1.30	0.89	0.60 to 1.32
Women	ref		ref	
Men	1.03	0.81 to 1.29	1.07	0.85 to 1.35
Expectations[b]	1.24	1.19 to 1.30	1.24	1.18 to 1.30
65 to 74 y (n = 3,777)	**Model 1 Dissatisfaction**		**Model 2 Dissatisfaction**	
	RR[a]	95% CI	RR[a]	95% CI
Preoperative anxiety or depression				
No	ref			
Yes	1.22	1.00 to 1.48		
Postoperative anxiety or depression				
No			ref	
Yes			2.61	2.15 to 3.17
ASA classification				
≤2	ref		ref	
≥3	1.28	1.00 to 1.61	1.24	0.98 to 1.56
Women	ref		ref	
Men	0.84	0.69 to 1.03	0.92	0.76 to 1.12
Expectations[b]	1.21	1.17 to 1.26	1.20	1.15 to 1.24
>74 y (n = 2,558)	**Model 1 Dissatisfaction**		**Model 2 Dissatisfaction**	
	RR[a]	95% CI	RR[a]	95% CI
Preoperative anxiety or depression				
No	ref			
Yes	1.34	1.07 to 1.69		
Postoperative anxiety or depression				
No			ref	
Yes			2.57	2.02 to 3.26
ASA classification				
≤2	ref		ref	
≥3	1.57	1.24 to 1.99	1.36	1.06 to 1.73
Women	ref		ref	
Men	0.93	0.73 to 1.17	0.93	0.74 to 1.18
Expectations[b]	1.21	1.16 to 1.27	1.20	1.14 to 1.26

The Exposure of Interest was Preoperative Anxiety/Depression in Model 1 and Postoperative Anxiety/Depression in Model 2.
ref, reference value; ASA, American Society of Anesthesiologists; RR, risk ratio; CI, confidence interval.
[a] Presented as risk ratio (RR) with 95% confidence interval (CI).
[b] For every 10 units increase in the visual analog scale ranging from 0 to 100 (high to low expectations).

be worried about going through a major surgery and not knowing what to expect rather than having pathological anxiety [19]. The translation for depression is the Swedish word "nedstämd", which translated back, means depressed mood and is one of many symptoms of clinical depression. Therefore, the results of this study should not be interpreted as indicating that 35% of Swedish patients who are scheduled for TKA have pathological anxiety or clinical depression. However, it does illustrate that it may benefit the patient if physicians address these symptoms, as they could indicate an illness requiring treatment or more attention.

The results of this study must be considered in the context of its strengths and potential limitations. The study includes a relatively large cohort with an acceptable response rate, including patients from several clinics (university, public, and private) across Sweden, which may be seen as a strength. Also, we consider the perspective of investigating those patients who report improvements in pain or function using the OMERACT-OARSI responder criteria as a strength. The respondent criteria consider both absolute and relative changes instead of using mean values. Thus, it provides insight into variables that may be important in a population that, from a surgical point of view, ought to be satisfied [29]. As for potential limitations, we want to highlight that we do not have information on whether any of the included patients were clinically diagnosed with anxiety or depression and treated for that. Furthermore, as missing PROM data are generally not considered to be missing at random, this could introduce bias in the study.

In conclusion, our findings indicate that preoperative and postoperative patient-reported anxiety or depression increase the risk of dissatisfaction after TKA despite improvements in pain or function. Anxiety and depression are potentially modifiable factors with psychological and pharmaceutical treatment and are therefore important for physicians to consider before and after a TKA in an effort to reduce the number of dissatisfied patients. Future studies comparing patients who have and do not have treatment for anxiety or depression or studies with interventions of patient empowerment or such, would add important knowledge and understanding of this area.

CRediT authorship contribution statement

Siri Heijbel: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Annette W-Dahl:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Josefine E-Naili:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Margareta Hedström:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization.

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